

Dinakar Murthy

Open to Relocate

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PROFESSIONAL SUMMARY

Data-driven Analyst skilled in predictive modeling, anomaly detection, and document understanding, with a track record of translating ML solutions into business value. Experienced in building and deploying models across domains like healthcare, HR, and marketing using Python, SQL, and cloud tools (AWS, Azure). Passionate about LLMs, agent-based simulation, and real-time analytics, with strong EDA, visualization (Tableau, QuickSight), and communication skills.

EDUCATION

San Francisco State University

Master of Science in Business Analytics

San Francisco, CA

May 2024

PES University

Bachelor of Technology in Computer Science and Engineering

Bengaluru, India

October 2021

TECHNICAL SKILLS

Programming Languages: Python, R, SQL

Software Tools and Packages: Tableau, Power BI, Pandas, NumPy, scikit-learn, Matplotlib, Seaborn, PyTorch, Mesa, PyTorch, OpenAI API, Streamlit, Flask, HTML/CSS, ReportLab, FAISS, Sentence Transformers, HuggingFace

Machine Learning & NLP: Natural Language Processing (NLP), Large Language Models (LLMs), Prompt Engineering, Text Classification, Resume Parsing, Multi-round Simulation, Semantic Search, Vector Embeddings, FAISS ANN Indexing

Cloud Technologies: AWS, Azure

System Design & Automation: Modular System Design, Workflow Orchestration, Evaluation Pipelines, Real-time Dashboarding, End-to-End AI System Integration, User-Interactive Dashboard with Filtering & Scoring

Leadership/Extracurricular: Project management, team leadership, and strategic planning, product ownership of AI-based solutions, cross-functional collaboration, and independent research-driven development, Technical storytelling and explainability through rich UI and documentation

PROFESSIONAL EXPERIENCE

ElevateMe Bootcamp

Data Analytics and Machine Learning Fellow

Columbus, OH

March 2025 - Present

- Utilized Python, Jupyter Notebooks, Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn to build and evaluate a machine learning risk assessment model for an event management company.
- Led collaboration across Sales, Marketing, and Finance to develop the predictive framework identifying early contract failure signals, enabling targeted interventions and reducing overall event risk exposure by 20%.

San Francisco State University

Lead Data Analyst & Researcher

San Francisco, CA

December 2022 - Present

- Enhanced search engine marketing (SEM) conversion rates by 40% by analyzing 400,000 keywords from Google Analytics, identifying trends, and recommending optimized strategies, through data-driven insights and strategic campaign adjustments.
- Developed self-service dashboards in Tableau to track campaign performance at campaign level with dynamic filtering to promote data driven decisions for campaign optimization strategies.

Delivered Korea

Lead Data Analyst

San Francisco, CA

July 2023 - May 2024

- Quantified the role of marketing interventions (influencer content, marketing spend and Google reviews) in boosting new customer acquisition and its effect on revenue, by identifying gaps in existing sales prediction models through in-depth research and developing an innovative sales forecasting model leveraging limited data to achieve an accuracy of 85%.
- Created clear visualizations (time series plots, correlation heatmaps) to illustrate trends and relationships between marketing spend, customer growth, and monthly sales, enhancing data-driven decision-making, by integrating data visualization tools and statistical analysis.

Kinetrix Technologies

Lead Data Analyst

Remote, India

June 2021 - July 2022

- Developed a social welfare web application using Django, integrating Python, SQL, Folium, and Google Maps API, enhancing user experience by 50%, through efficient data integration and UI design.

- Improved project efficiency as Project Manager by employing best practices in data analysis and technology integration, resulting in 20% reduction in project timelines, by streamlining workflows and enhancing team productivity.

Jainstein

Bangalore, India

Data & Research Intern

May 2019 - June 2020

- Authored and edited 27,000+ words of original content, including introduction and literature review chapters, using data-backed arguments and structured research synthesis across various domains.
- Interpreted qualitative and quantitative research data to draft compelling narratives that aligned with academic standards, supporting broader research objectives with clarity and precision.
- Adhered to strict data privacy protocols, maintaining intellectual property confidentiality and securely managing sensitive research material in accordance with zero-tolerance data breach policy.

PROJECTS

AI-Powered Semantic Product Search Engine | [GitHub](#)

April 2025 - May 2025

- Spearheaded the end-to-end development of an AI-powered semantic product search engine for fashion e-commerce, leveraging MiniLM embeddings and FAISS vector indexing to improve search relevance beyond keyword matching by 3x in test cases.
- Engineered an interactive Streamlit UI with real-time filtering, image-based product cards, and hover-zoom UX, enabling users to query and visualize ~44K products with sub-second latency on a local M2-powered environment.

AI Interview Analysis Sandbox | [GitHub](#)

April 2025 - April 2025

- Built a full-stack AI-powered interview simulation system that dynamically generates, conducts, and evaluates multi-round interviews using LLMs—integrating resume parsing, contextual question generation, and automated scoring into a scalable, modular Python pipeline.
- Incorporated multi-turn conversation logic and feedback loops to enhance realism and provide constructive, scenario-based interview simulations
- Conducted iterative prototyping and user testing to refine question logic, model prompt design, and session flow for realistic interview pacing.

Stock Market Simulation Engine | [GitHub](#)

April 2025 - April 2025

- Developed a simulation tool using Python to model investor behavior and market dynamics, allowing users to test and visualize stock investment strategies under randomized market conditions.
- Implemented object-oriented principles to simulate market agents, stock volatility, and profit/loss tracking; incorporated data visualization (Matplotlib) for time-series analysis of investment performance.

Virus Spread Simulator | [GitHub](#)

April 2025 - April 2025

- Engineered a Python/Mesa simulation core that drives 10 k-agent grids at interactive speed, modeling probabilistic infection, recovery, fatality, immunity waning, reinfection, and vaccination campaigns with tunable efficacy and start-time parameters.
- Exposed a clean configuration layer & batch-runner hooks so researchers can sweep over key epidemiological levers (β , γ , asymptomatic fraction, mobility toggles) and export Mesa DataCollector outputs to Pandas for downstream analysis of intervention scenarios.

Conway's Game of Life Engine & ML Validator | [GitHub](#)

March 2025 - April 2025

- Implemented Conway's Game of Life using Python, simulating cellular automata and emergent behavior from simple rules and initial conditions; Designed and optimized the simulation logic to handle grid state transitions efficiently, applying rule-based updates across multiple iterations.
- Built reusable components for simulation logic, enabling parameter tuning and extension into other automata-based simulations
- Designed and trained a self-supervised 3D convolutional neural network in PyTorch to validate simulation physics—including wormhole behavior—by computing per-frame binary cross-entropy error to flag logic deviations in real time.

AI Chat Bot | [GitHub](#)

January 2024 - May 2024

- Developed a full-stack AI-powered chat application using the MERN stack, integrating real-time communication with Socket.IO and secure JWT-based authentication.
- Engineered backend services to manage user sessions, prompt handling, and API interaction with OpenAI for dynamic response generation.
- Built a responsive React front-end for intuitive real-time chatting with bot and peer users, using WebSockets for seamless message updates

OTHER ACHIEVEMENTS

Master of Science in Business Analytics Program - San Francisco State University

San Francisco, CA

Academic Excellence Award for outstanding performance

Data Analytics and Machine Learning Certificate – ElevateMe Bootcamp

Columbus, OH